

Part I

SUMMARY of Solving Optimization Problems

Summary

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0.1 How to Solve an Optimization Problem

- (a) Identify **variables**; draw and label the picture of the problem
- (b) Identify the **objective function** (a quantity to be optimized);
write a **formula** for the objective function in terms of variables of the problem
- (c) Identify the **constraint(s)**;
use the constraint(s) to express all the variables in terms of a **single variable**
- (d) Write the **objective function in terms of a single variable**;
find the **interval of interest**
- (e) Using the methods of calculus, find the **global maximum/minimum**;
justify your answer

REMINDER: The Interval of Interest tells you the method to use to solve an optimization problem.

CASE 1: The interval of interest is a closed interval $[a, b]$

In this case, the Extreme Value Theorem (EVT) guarantees that both global extrema of f exist!

If f has an global minimum and an global maximum which either occur at a critical point or at a boundary point (which means a or b). To find an extreme values of f on $[a, b]$, we:

- Find all critical points and plug them into f .
- Evaluate f at both boundary points.
- Compare the values. The biggest of those values is the maximum value of f on $[a, b]$, and the least one is the minimum value of f on $[a, b]$.

CASE 2: The interval of interest is any interval.

In this case, the global minimum/maximum **may not exist**. Our approach is then:

- Find all critical points of f on the interval.
- Use 1st or 2nd Derivative Test to classify those critical points as **local maxima/minima**.

Summary

- If there is *exactly one local extremum*, then it is an global extremum of the same type. (That is, if it is a local minimum then it is automatically a global minimum...)

OTHER CASES (e.g., An interval with multiple local extrema, etc) can be approached similar to how we graphed functions: check the boundary points, check the critical points, use the sign chart etc.

Part II

Recitation Questions

Problem 1

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Problem 1

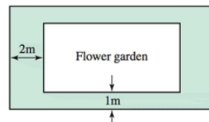
Problem 1 Suppose you want to maximize a continuous function on a closed interval, but you find that it only has one local extremum on the interval which happens to be a local minimum. Where else should you check for the solution? **EXPLAIN.**

Problem 2

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Problem

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(see figure)
garden



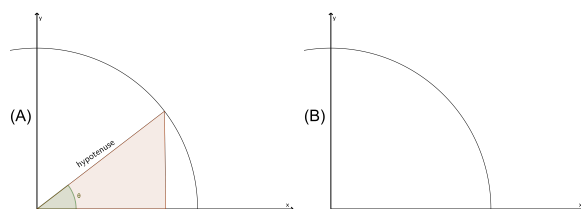
- (a) Label the picture with variables.
- (b) What are you trying to maximize or minimize? Write an equation for it in terms of the variables from (a).
- (c) What is your constraint? Write a constraint equation in terms of the variables from (a).
- (d) Reduce your optimization equation to one variable using the constraint equation.
[2]
- (e) What is the interval on which your variable makes sense? Is it open or closed? What does this mean for the method of finding the global max or min? [2]
- (f) Use the appropriate method to find and justify your global extremum. [4]
- (g) Be sure to answer the question asked in the original problem.

Problem 3

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Problem 3

Problem 3 A part of a circle centered at the origin with radius $r = 7$ cm is given in the figure (A) below. A right triangle is formed in the first quadrant (see figure (A)). One of its sides lies on the x -axis. Its hypotenuse runs from the origin to a point on the circle. The hypotenuse makes an angle θ with the x -axis.



Make sure to label the picture.

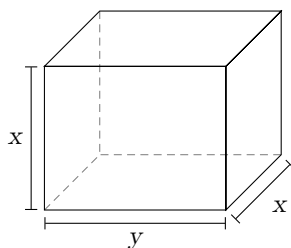
- Draw 2 more examples of such a triangle in the figure (B).
- Express the area of such a triangle as a function of θ and state its domain.
- Find the value of θ which maximizes the area in part (b). Show your work and justify your answer. [2]

Problem 4

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Problem 4

Problem 4 A box with two square sides is constructed to have a volume of 27. Denote the dimensions of the length of the base as y , and the width and height as x , as labeled in the diagram below. Find the dimensions needed to minimize the surface area of the box.



Solve the problem by performing the following steps.

- (a) Find a formula for $S(x)$ the surface area of the box, as a function of only the variable x .
- (b) Find the interval of interest for $S(x)$.
- (c) Find the x -value that gives the minimum surface area over all such boxes.

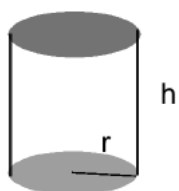
[2]

Problem 5

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Problem 5

Problem 5 Find the radius of a cylindrical container with a volume of $2\pi \text{ m}^3$ that minimizes the surface area.



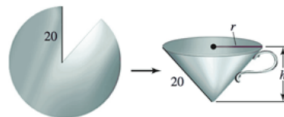
(HINT: surface area , $S = 2\pi rh + 2r^2\pi$;
volume , $V = r^2\pi h$.)

Problem 6

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Problem 6

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Problem 7

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Problem 7

Problem 7 What point on the parabola $y = 5 - x^2$ is closest to the point $(4, 7)$?

Problem 8

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Problem 8

Problem 8 *A rectangle is constructed with one side on the positive x -axis, one side on the positive y -axis, and the vertex opposite the origin on the line $y = 10 - 2x$. What dimensions maximize the area of the rectangle? What is the maximum area?*

Problem 9

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Problem 9

Problem 9 Suppose you own a tour bus and you book groups of 20 to 80 people for a day tour. The cost per person is \$30 minus \$0.25 for every ticket sold. If gas and other miscellaneous costs are \$300, how many tickets should you sell to maximize your profit? Treat the number of tickets as a nonnegative real number.